

MedCare Network

Hospital Operations & Revenue Analysis

Multi-Branch Healthcare Dataset - Exploratory Data Analysis Report

Prepared for: Healthcare Analytics Portfolio | July 2026

Dataset: 5 linked tables (doctors, patients, appointments, treatments, billing)

Branches Covered: Central Hospital, Eastside Clinic, Westside Clinic

Note: This dataset is a small, deliberately messy sample (200 appointment records). Findings illustrate the analytical process and directional patterns rather than statistically definitive conclusions. See the Data Disclaimer in Section 15 for full context.

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Section 1 - Introduction & Objective

Introduction

MedCare Network operates three hospital branches: Central Hospital, Eastside Clinic, and Westside Clinic, spanning three specializations: Pediatrics, Dermatology, and Oncology. This report presents a complete Exploratory Data Analysis (EDA) of the network's operational dataset, covering 200 cleaned appointment records linked across doctors, patients, treatments, and billing.

Objective

- Understand appointment outcomes, no-show behaviour, and where operational load is concentrated across branches and specializations.
- Analyze treatment costs and billing patterns to identify where revenue is collected, delayed, or lost.
- Compare doctor staffing levels against patient demand by specialization.
- Track appointment volume over time to spot seasonal or monthly patterns.
- Surface real-world operational problems from the data and propose actionable, data-backed solutions.

Tools & Environment

Tool/Library	Purpose
Python 3	Core programming language
Polars	Data cleaning and transformation
Pandas & Numpy	Aggregation, statistics, and insight generation
Matplotlib & Seaborn	Statistical visualization
Plotly	Interactive chart prototyping
VS Code	Code editor and development environment

Section 2 - Dataset Overview

The dataset consists of 5 linked CSV files representing a small hospital network: doctors, patients, appointments, treatments, and billing. It was deliberately kept messy (inconsistent casing, mixed date formats, typo'd status labels, and a small number of duplicate and orphaned records) to simulate a realistic operational export.

Raw vs. Cleaned Summary (Across All 5 Tables)

Metric	Raw Dataset	Cleaned Dataset
Total Rows (all tables combined)	670	660
Total Tables	5	5
Total Nulls (across all tables)	109	0
Duplicate Rows Found	10	0
Rows Dropped	-	10 (confirmed duplicates)
Orphaned Foreign Keys	Present (e.g. doctor_id 'D999')	Nullled, not guessed

Table-by-Table Row Counts

Table	Raw Rows	Cleaned Rows	Rows Dropped	Reason
doctors	10	10	0	No duplicates or invalid rows found
patients	53	50	3	Exact duplicate patient records
treatments	200	200	0	8 missing cost/type values imputed, not dropped
appointments	204	200	4	Exact duplicate appointment records
billing	203	200	3	Exact duplicate billing records

Column Description (Key Tables)

Column	Table	Nulls (Raw)	Description
doctor_id / patient_id	doctors / patients	0	Primary identifiers used to link records across tables
specialization	doctors	0	Pediatrics, Dermatology, or Oncology
years_experience	doctors	0	Sometimes stored with stray text (e.g. '17 yrs') - needed cleaning
gender	patients	0	Stored inconsistently as M/F/male/female - standardized
date_of_birth	patients	0	Mixed date formats; some future or 110+ year ages found and nulled
status	appointments	0	Scheduled, Completed, Cancelled, No-show - typos and casing fixed
cost	treatments	8	Sometimes included currency symbols and commas
amount	billing	0	Occasionally did not match the linked treatment cost
payment_status	billing	0	Paid, Pending, or Failed

Section 3 - Data Cleaning - Steps & Decisions

Every cleaning decision was made based on data evidence, not assumptions. Each step is documented with the reason behind it.

Step	Action	Reason
1	drop_duplicates() on all 5 tables	10 exact duplicate rows found and removed
2	Standardize text casing (names, specialization, insurance provider)	Fixed mixed casing such as 'dermatology' vs 'DERMATOLOGY'
3	gender: map M/F/male/female variants	Normalized to two canonical values: Male, Female

Step	Action	Reason
4	status: map typos/casing to 4 categories	e.g. 'No Show', 'NoShow', 'no-show' all mapped to 'No-show'
5	payment_status: map typos/casing	e.g. 'paid', 'PAID' mapped to 'Paid'
6	date_of_birth / dates: parse 5 known formats	Raw dates mixed YYYY-MM-DD, DD-MM-YYYY, MM/DD/YYYY, and others
7	date_of_birth: null out invalid ages	Future dates and ages above 110 years could not be trusted - nulled rather than guessed
8	years_experience: strip non-numeric text	Raw values like '17 yrs' converted to integer 17
9	cost / amount: strip symbols and commas	Raw values like 'Rs. 3,941.97' cleaned before numeric conversion
10	cost: abs() on negative values	Negative costs found in raw data - treated as sign entry errors, not dropped
11	cost: fill missing via median per treatment_type	8 missing costs filled using the median for that treatment type, with an overall median fallback
12	contact_number: strip to digits, trim country code	Raw numbers had dashes, spaces, brackets, and +91 prefixes
13	Missing email/address/phone: fill with 'MISSING'	Cannot be guessed - flagged clearly instead of left blank or faked
14	doctor_id: validate against known doctors table	Invalid IDs such as 'D999' nulled rather than kept as a broken link
15	reason_for_visit: normalize blanks and 'N/A'	Both mean the same thing - unified into 'Not Specified'
16	billing.amount: reconciled against treatment.cost	Amount corrected to match the linked treatment cost wherever the two disagreed
17	Memory optimization: downcast numerics, categorical dtypes	Reduced memory footprint for low-cardinality text columns

Section 4 - Exploratory Data Analysis

Key Metrics Summary

Appointments Analyzed	Overall No-show Rate	Average Treatment Cost	Total Billed Revenue
200	26.00%	Rs. 2,738.91	Rs. 4,85,224.31

Patient Age Distribution

Metric	Value
Min	21.2 years
25th %ile	32.0 years
Median	39.9 years
Mean	43.9 years

Metric	Value
75th %ile	56.1 years
Max	76.5 years
Std Dev	16.2 years

Treatment Cost Distribution

Metric	Value
Min	Rs. 534.03
25th %ile	Rs. 1,640.54
Median	Rs. 2,769.60
Mean	Rs. 2,738.91
75th %ile	Rs. 3,729.78
Max	Rs. 4,973.63
Std Dev	Rs. 1,267.55

Appointment Status Breakdown

Status	Count	% Share
No-show	52	26.0%
Scheduled	51	25.5%
Cancelled	51	25.5%
Completed	46	23.0%

Reason for Visit

Reason	Count
Therapy	38
Consultation	37
Checkup	36
Follow-up	35
Not Specified	28
Emergency	26

Note: 'Not Specified' (28 records) combines genuinely blank entries and the literal text 'N/A' found in the raw data, unified during cleaning.

Section 5 - Visualization 1 - Hospital Overview Dashboard

Hospital Overview Dashboard — Operations (top) & Finance (bottom)

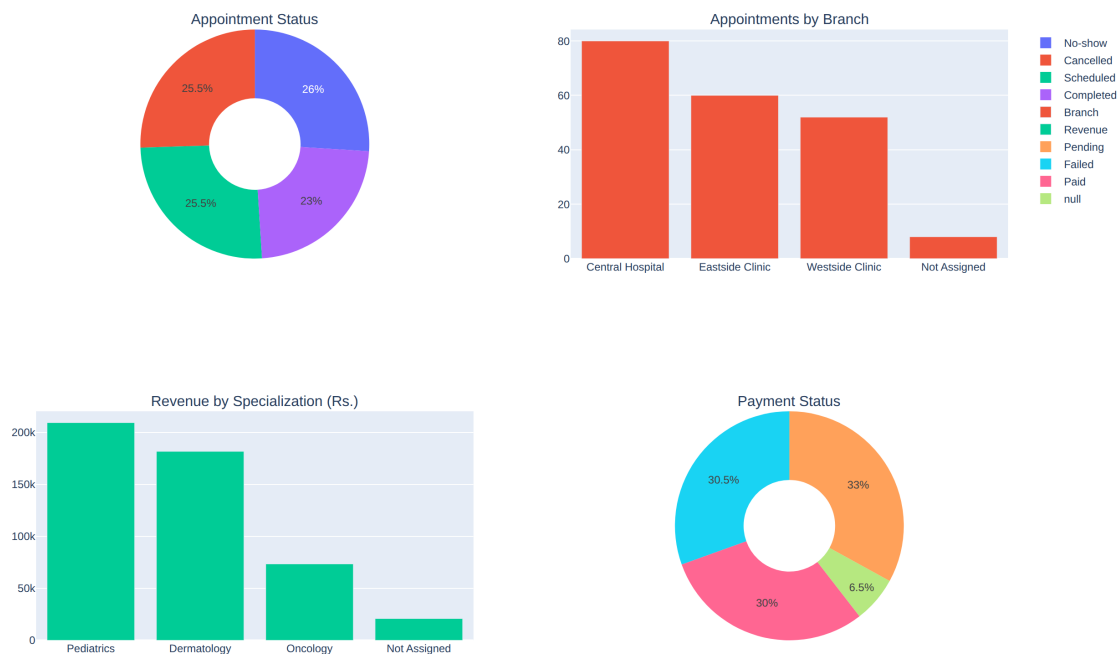


Fig 1. Four-panel dashboard: appointment status (top-left), appointments by branch (top-right), revenue by specialization (bottom-left), and payment status (bottom-right).

What does this chart mean?

This dashboard puts operations (top row) next to finance (bottom row) so the two can be read together. It answers a simple operational question: which branch handles the most traffic, how much of that traffic actually completes as a visit, and is the resulting revenue actually being collected?

Key Business Takeaways

- **No-shows Are the Single Largest Outcome:** At 26%, no-shows outnumber every other single status, including completed visits (23%). More appointments are lost to absence than are actually fulfilled.
- **Central Hospital Carries the Heaviest Load:** Central Hospital handles 80 of 200 appointments (40%), more than Eastside Clinic (60) or Westside Clinic (52) individually, making it the single largest concentration of operational load in the network.
- **Pediatrics Leads on Revenue, Pending Payments Lead on Volume:** Pediatrics generates the highest revenue by specialization, while on the payment side, Pending is the single largest bucket, ahead of both Paid and Failed individually - more money is uncollected than collected.

Section 6 - Visualization 2 - Specialization to Doctor to Treatment Type

Specialization -> Doctor -> Treatment Type

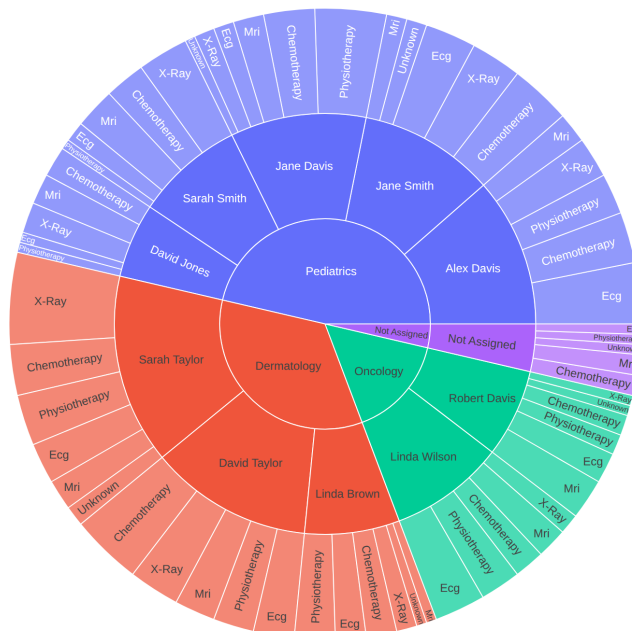


Fig 2. Sunburst chart: inner ring is specialization, middle ring is individual doctor, outer ring is treatment type.

What does this chart mean?

This sunburst breaks the network down in three layers: specialization first, then the individual doctor within it, then the treatment types that doctor performed. The size of each slice is proportional to appointment count, so the biggest slices show exactly where the network's caseload is concentrated, down to the individual doctor.

Key Business Takeaways

- **Pediatrics Is the Largest Inner Ring:** 5 Pediatrics doctors handle 94 appointments between them, the largest specialization segment, and the ring is split fairly evenly across all five doctors.
- **Dermatology's Load Sits on Fewer Doctors:** 68 Dermatology appointments are split across just 3 doctors (22.7 per doctor on average), the highest per-doctor share of the three specializations, visible as noticeably larger middle-ring segments than Pediatrics.
- **Treatment Mix Varies by Doctor, Not Just by Specialty:** The outer ring shows that individual doctors within the same specialization lean toward different treatment types, useful for spotting which doctor to route a specific treatment request to.

Section 7 - Visualization 3 - Treatment Cost Distribution by Type

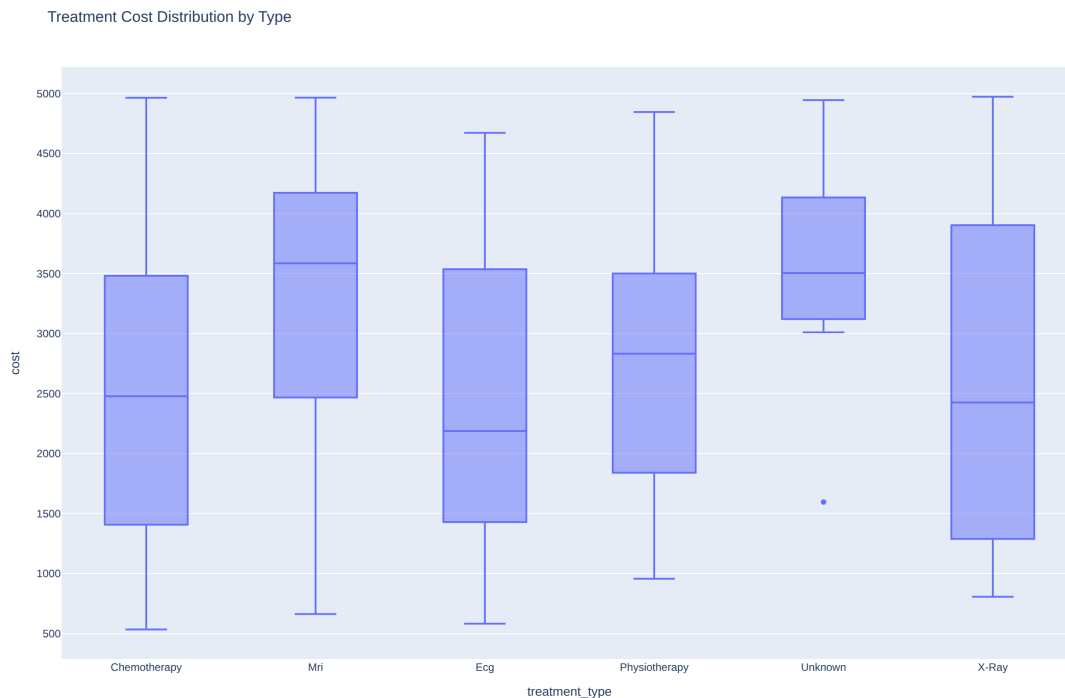


Fig 3. Box plot of treatment cost by treatment type, showing median, interquartile range, and outliers for each category.

What does this chart mean?

This box plot shows not just the average cost per treatment type, but how much that cost varies within each type. A tall box means a treatment type is priced inconsistently; a short box means it is billed fairly predictably.

Key Business Takeaways

- **MRI Is Both the Costliest and the Widest-Ranging:** MRI has the highest median cost (Rs. 3,585) and one of the widest spreads (Rs. 663 to Rs. 4,966), making it the hardest treatment type to estimate a bill for in advance.
- **Unknown Treatment Type Clusters at the High End:** Records where treatment_type was missing in the raw data (labelled 'Unknown') show a median of Rs. 3,504, close to MRI's, reinforcing that this is likely a data-entry gap on genuinely costlier procedures rather than a distinct low-cost category.
- **Chemotherapy and ECG Are the Most Predictably Priced:** Chemotherapy (median Rs. 2,478) and ECG (median Rs. 2,188) show the most compact boxes, meaning their billing is more consistent and easier to forecast than MRI or X-Ray.

Section 8 - Visualization 4 - Monthly Appointment Trend

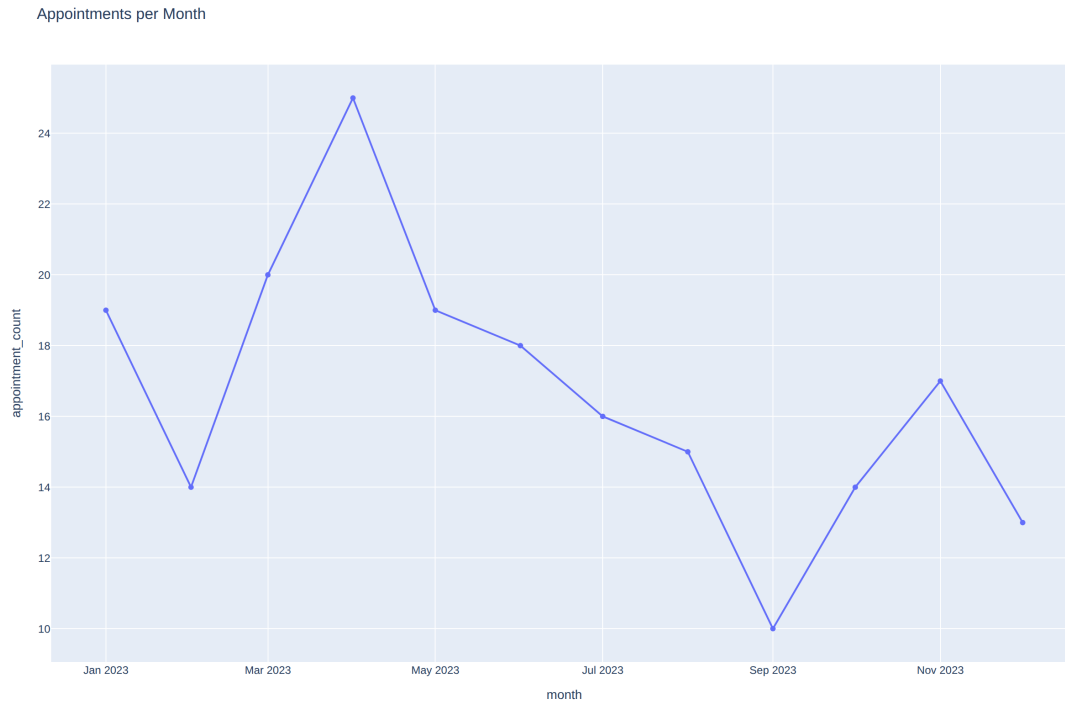


Fig 4. Appointment volume by month across 2023.

What does this chart mean?

This line chart tracks how many appointments were booked each month across the year. It helps answer a scheduling and staffing question: are there predictable busy and quiet periods the network should plan around?

Key Business Takeaways

- **April Is the Peak Month:** April recorded 25 appointments, the highest of any month, two and a half times the 10 appointments recorded in the quietest month.
- **September Is the Quietest Month:** September dropped to just 10 appointments, the lowest point in the year, a dip worth investigating alongside staff leave schedules or seasonal illness patterns.
- **A Gradual Cooling Trend After Mid-Year:** Volume trends generally downward from the April peak through the second half of the year, before a modest recovery in November.

Section 9 - Visualization 5 - No-show Rate by Specialization

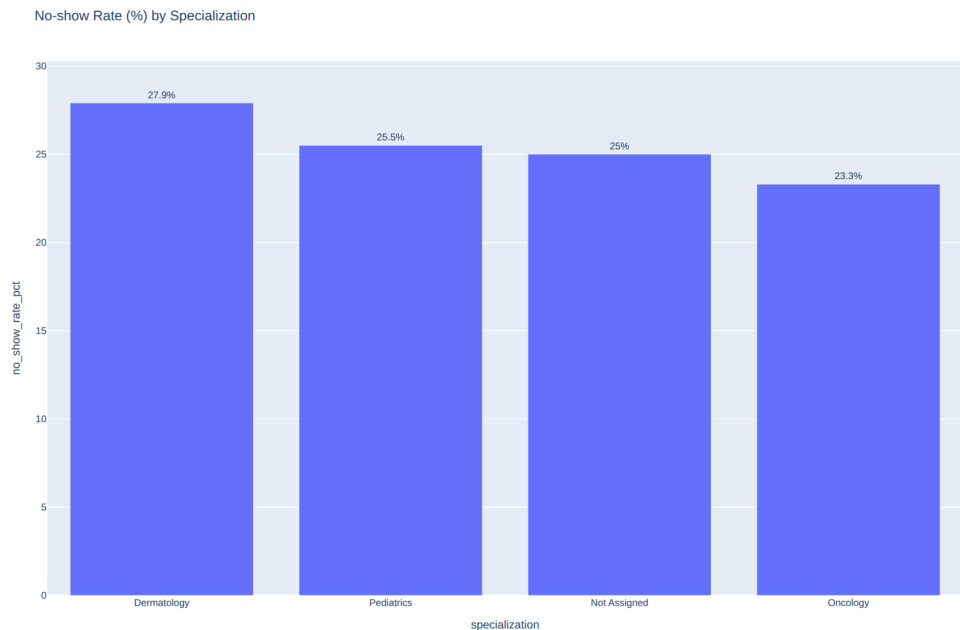


Fig 5. Percentage of appointments ending in no-show, by specialization, sorted highest to lowest.

What does this chart mean?

This chart isolates the no-show problem by specialization to identify where patient attendance is weakest, rather than treating the 26% network-wide rate as one uniform number.

Key Business Takeaways

- **Dermatology Has the Worst Attendance:** Dermatology no-shows run at 27.94%, above the network average, and is also the specialization with the widest experience gap between its doctors.
- **Oncology Patients Are the Most Reliable:** Oncology no-shows sit at just 23.33%, the lowest of the three, plausibly because oncology visits carry higher perceived urgency.
- **Gender Also Plays a Small Role:** Male patients no-show slightly more often than female patients (26.92% vs 24.29%), a smaller effect than the specialization gap but worth tracking alongside it.

Section 10 - Branch & Revenue Analysis

Revenue is not evenly distributed across the network's three branches or three specializations. Understanding where money is actually generated helps prioritize where operational fixes will have the biggest financial impact.

Revenue by Branch

Branch	Appointments	Revenue (Rs.)	Observation
Central Hospital	80	1,95,862.74	Highest revenue and highest volume - the network's primary hub
Westside Clinic	52	1,43,244.54	Fewer appointments than Eastside but higher revenue, suggesting costlier treatment mix

Branch	Appointments	Revenue (Rs.)	Observation
Eastside Clinic	60	1,25,222.63	Highest appointment count of the two smaller branches, but lowest revenue

Revenue by Specialization

Specialization	Revenue (Rs.)	Doctors	Revenue per Doctor (Rs.)
Pediatrics	2,09,270.70	5	41,854.14
Dermatology	1,81,651.72	3	60,550.57
Oncology	73,407.49	2	36,703.75

Dermatology generates the highest revenue per doctor despite having fewer appointments than Pediatrics, driven by its individually costlier procedures.

Section 11 - Payment & Billing Analysis

Payment Status	Count	Total Billed (Rs.)	Average Treatment Cost (Rs.)
Pending	66	1,67,147.63	2,652.39
Failed	61	1,69,780.24	2,863.76
Paid	60	1,48,296.44	2,703.63

Key Business Takeaways

- **Failed Payments Skew Toward Costlier Treatments:** Failed payments average Rs. 2,863.76 in treatment cost versus Rs. 2,676.79 for paid or pending combined - higher bills appear to be genuinely harder to collect.
- **Pending Is the Largest Single Bucket:** 66 of 200 bills sit in Pending status, more than either Paid or Failed individually, representing collectible but currently uncollected revenue.
- **Age Has Almost No Relationship With Treatment Cost:** The correlation between patient age and treatment cost is just 0.07, effectively no link - billing risk should be assessed by treatment type and payment method, not patient demographics.

Section 12 - Real-World Problems & Solutions

Based on the data patterns analyzed above, the network faces specific operational challenges. Here are the core problems and data-backed solutions.

Problem 1: No-shows Outnumber Completed Visits

- **The Data:** 26% of all appointments end in no-show, more than the 23% that are actually completed. Every fourth booked slot is being wasted.
- **The Solution:** Introduce automated appointment reminders (SMS or call) 24 to 48 hours in advance, prioritizing Dermatology, which has the highest no-show rate at 27.94%. A confirmation step before the visit is the fastest lever to recover lost slots.

Problem 2: Rs. 1.69 Lakh in Failed Payments Are Concentrated in Costlier Treatments

- **The Data:** Failed payments average Rs. 2,863.76 per treatment, noticeably higher than paid or pending bills, and total Rs. 1,69,780.24 across 61 records.
- **The Solution:** Offer installment or partial-payment options specifically for treatments above the Rs. 2,769.60 median cost, rather than requiring full payment upfront, to reduce the failure rate on the network's most expensive procedures.

Problem 3: Staffing Is Not Matched to Demand

- **The Data:** Dermatology doctors handle 22.7 appointments each on average, the highest per-doctor ratio in the network, and one of the three Dermatology doctors (D008) has just 5 years of experience against a 17-26 year range for the rest of the team, meaning the most heavily loaded specialization also carries its least experienced doctor.
- **The Solution:** Redistribute a share of Dermatology's caseload to reduce its 22.7-per-doctor load toward the network average, and pair the newer Dermatology doctor (D008) with a senior colleague on complex cases until outcomes are better understood.

Problem 4: Eight Appointments Have No Assigned Doctor

- **The Data:** 8 appointments carry an invalid or missing doctor_id, meaning they cannot be attributed to any branch or specialization for reporting purposes.
- **The Solution:** Add a mandatory doctor-assignment validation step at the point of booking, rather than catching the gap later in analysis, to keep every appointment reportable.

Section 13 - Recommendations

- **Prioritize a No-show Reduction Program:** Roll out reminder calls or texts network-wide, starting with Dermatology, to directly target the largest single source of lost appointments.
- **Introduce Flexible Billing for High-Cost Treatments:** Offer installment plans for treatments above the median cost to reduce the Rs. 1.69 lakh currently sitting in failed payments.
- **Rebalance Doctor Workload:** Use appointments-per-doctor as a standing operational metric, not just a one-time finding, to catch staffing imbalances before they affect wait times.
- **Close the Data Entry Gap at Booking:** Enforce doctor assignment and standardized status labels at the point of entry so future reporting does not need to reconstruct missing links after the fact.
- **Investigate the September Dip:** Cross-check the September appointment slowdown against staff leave calendars and local seasonal factors to determine whether it is structural or one-off.

Section 14 - Technical Challenges & Problem Solving

Multi-Table Relational Cleaning

Unlike a single flat file, this dataset required cleaning 5 interdependent tables in the correct order, patients and doctors before appointments, and treatments before billing, so that foreign key checks and cost reconciliation had a trustworthy source table to validate against.

Inconsistent Date Formats

Dates appeared in at least 5 different formats across the raw files (YYYY-MM-DD, DD-MM-YYYY, MM/DD/YYYY, YYYY/MM/DD, and DD/MM/YYYY). Each date column was parsed by attempting all known formats and keeping the first successful match, rather than assuming one fixed format.

Cost vs. Billing Amount Reconciliation

The billing table's amount did not always match the linked treatment's cost. Rather than trusting either value blindly, the treatment cost was treated as the source of truth and used to correct any mismatched billing amount.

Orphaned Foreign Keys

Some appointments referenced doctor IDs (such as 'D999') that did not exist in the cleaned doctors table. These were nulled out rather than kept as broken links, so downstream aggregations by specialization and branch would not silently misattribute them.

Smart Imputation Over Blind Deletion

8 missing treatment costs were filled using the median cost for that specific treatment type, with an overall median as a fallback, instead of dropping the rows or using one flat network-wide average that would have ignored the real cost differences between, say, an ECG and an MRI.

Section 15 - Conclusion

This Exploratory Data Analysis of the MedCare Network dataset transforms 200 appointment records across 5 linked tables into actionable operational intelligence. The findings show that the network's biggest opportunities are not in acquiring new patients, but in reducing no-shows, tightening billing collection on high-cost treatments, and matching doctor staffing more closely to real patient demand.

By executing this project end to end, from multi-table relational cleaning and foreign key validation to statistical visualization and revenue analysis, this report demonstrates a complete analytical workflow: turning a small, messy, realistic healthcare export into a clear set of operational recommendations.

Data Disclaimer & Real-World Constraints

This is a small, deliberately messy sample dataset (200 appointment records, 10 doctors, 50 patients). It was built to demonstrate a realistic cleaning and analysis workflow rather than to represent a statistically robust sample of any real hospital network. Percentages and averages calculated on small subgroups, such as the 30 Oncology appointments, should be read as directional patterns rather than precise population estimates.

Every cleaning decision in this report was made based on visible evidence in the data (duplicate rows, invalid dates, broken foreign keys, mismatched billing amounts) rather than assumptions, and flagged rather than silently discarded wherever a value could not be reliably reconstructed. True data cleaning preserves as much business value as possible while handling inconsistencies logically, rather than aggressively deleting anything imperfect to reach a falsely tidy table.